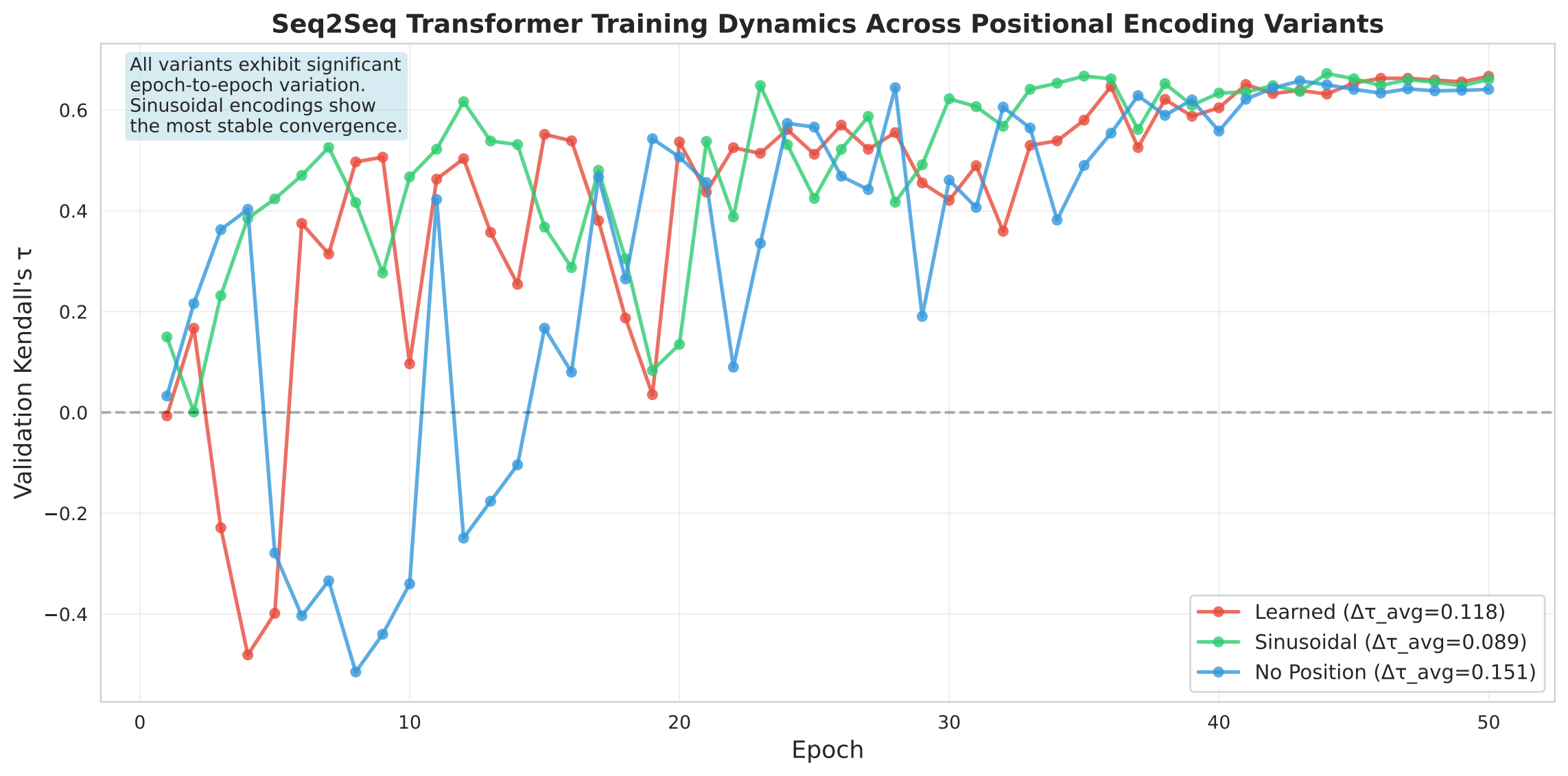
Seq2Seq Transformer Training Dynamics Across Positional Encoding Variants
All variants exhibit significant epoch-to-epoch variation. Sinusoidal encodings show the most stable convergence.
Validation Kendall's τ
0.6
0.4
0.2
0.0
−0.2
−0.4
0
10
20
30
40
50
Epoch
Learned (Δτ_avg=0.118)
Sinusoidal (Δτ_avg=0.089)
No Position (Δτ_avg=0.151)